

Software Test Plan (STP) - Vehicle Parking System

Project: Vehicle Parking System (VPS)

Version: 1.0

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Status: Submitted for Part-1 review; no test has been executed yet

Course: Software Engineering (UE24CS341A)

1. Introduction

Purpose: This document defines how release 1.0 of the Vehicle Parking System will be tested: what is tested, how, by whom, with what tools and data, and how results are traced back to the requirements in the SRS.

Scope: All functional requirements (FR), non-functional requirements (NFR), safety requirements (SAF) and security requirements (SEC) in SRS version 1.0, for the Django web application, the scheduler jobs and the forecast service. Items outside scope are listed in Section 4.

Objectives:

- Show that every SRS requirement has at least one test case or, where a test is not practical, a stated verification method.
- Prove the safety-critical rule that no slot is ever double-booked, including under concurrent requests.
- Validate the security requirements with both automated tests and scanning tools (Section 5.1).
- Keep regressions out of the main branch by running the automated suites in CI on every pull request.

References: VPS SRS version 1.0; VPS Software Architecture and Design Specification version 1.0; IEEE Std 829-2008 (Software and System Test Documentation); OWASP Web Security Testing Guide 4.2.

2. Test Items

Items under test are built from the main branch of the project repository; each test run records the commit hash.

Item	Version	Notes
Accounts and Access module (vps/accounts)	1.0	Registration, sign-in, logout, reset, roles
Vehicles module (vps/vehicles)	1.0	Registry and number validation

Item	Version	Notes
Lots and Slots module (vps/lots)	1.0	Lots, slots, blocks, moving bookings
Reservations and Slot Allocation modules	1.0	State machine, rules, allocation, expiry job
Gate Operations and Billing modules	1.0	Check-in, check-out, walk-in, charges, payments
Reports and Audit Log modules	1.0	Reports, CSV export, audit search
Web pages (templates and HTMX)	1.0	All Driver, Attendant and Administrator screens
Forecast service (forecast/app)	1.0	Forecast API and fallback behaviour
Database schema and migrations	1.0	Constraints, including the no-overlap exclusion constraint

3. Features to be Tested

Features are mapped to SRS requirement IDs and to the test cases in Section 16.

Feature (priority)	SRS requirements	Test cases
5.1 User Accounts and Access Control (High)	FR-AUTH-01 to FR-AUTH-06	TC-01, TC-02, TC-03, TC-21, TC-22, TC-23
5.2 Vehicle Registry (High)	FR-VEH-01 to FR-VEH-04	TC-04, TC-24
5.3 Lot and Slot Management (High)	FR-LOT-01 to FR-LOT-04	TC-05, TC-25, TC-26
5.4 Slot Availability Search (High)	FR-SRCH-01 to FR-SRCH-03	TC-05, TC-27, TC-28
5.5 Reservation Management (High)	FR-RES-01 to FR-RES-07	TC-06, TC-07, TC-08, TC-09, TC-10, TC-26, TC-29
5.6 Slot Allocation (High)	FR-ALLOC-01 to FR-ALLOC-03	TC-06, TC-07, TC-23
5.7 Gate Check-in and Check-out (High)	FR-GATE-01 to FR-GATE-07	TC-11, TC-12, TC-13, TC-30, TC-31, TC-32
5.8 Pricing and Billing (High)	FR-BILL-01 to FR-BILL-05	TC-13, TC-14, TC-30, TC-33
5.9 Reports and Audit (Medium)	FR-RPT-01 to FR-RPT-04	TC-19, TC-28, TC-34
5.10 Occupancy Forecast (Low)	FR-FCST-01 to FR-FCST-03	TC-20, TC-35
Performance (SRS 6.1)	NFR-PERF-01 to NFR-PERF-05	TC-18, TC-36, TC-37
Safety (SRS 6.2)	SAF-01 to SAF-03	TC-05, TC-23, TC-31
Security (SRS 6.3)	SEC-01 to SEC-10	TC-03, TC-15, TC-16, TC-17, TC-19, TC-22, TC-31, TC-38, TC-39, TC-40
Quality attributes (SRS 6.4)	NFR-REL, NFR-USE, NFR-MNT, NFR-PRT, NFR-TST	TC-07, TC-41, TC-42, TC-43

4. Features Not to be Tested

Item	Reason
Payment gateway or card	Payments are recorded by attendants only; no gateway is

Item	Reason
processing	integrated.
Physical gate hardware and number-plate cameras	Not part of this release.
Accuracy of the forecast model	FR-FCST is advisory and Low priority; only availability and fallback (TC-20) are tested.
SMTP provider delivery guarantees	E-mails are tested up to hand-off to the mail server; delivery is the provider's responsibility.
Browser vendor bugs	Cross-browser checks cover the versions in NFR-PRT-01 only.

NFR-AVL-01 (99% availability) is not verified by a test case; it is measured by uptime monitoring once the system is deployed (Section 14).

5. Test Approach / Strategy

Levels:

Level	What is tested	Tools	Run when	Lead
Unit	Service functions: allocation, billing, validation, state transitions; time passed in as a parameter	pytest, pytest-django, freezegun	Every push (CI)	Gitanjali
Integration	Views and APIs with a real PostgreSQL database; constraints, permissions, concurrency, e-mail outbox	pytest with a PostgreSQL 16 service container, DRF test client, thread pool	Every push (CI)	Hari
System	End-to-end flows in a browser; performance; security scans	Playwright, Locust, OWASP ZAP	Before each deliverable and before merging release branches	Dhruv (UI), Vivian (performance, scans)
Acceptance	Usability with first-time users; demo scenarios	Moderated sessions, timer, checklist	Before the final demo	Whole team

Types: functional, regression (the full automated suite on every pull request), concurrency (TC-07), performance and load (Locust), security (Section 5.1), usability and accessibility (axe-core), and cross-browser compatibility (Playwright on Chromium, Firefox and WebKit).

Test data: factory functions create users, lots, slots and reservations for each test; a seeded synthetic data set of one 500-slot lot with 30 days of reservations is used for performance tests. No real personal data is used. Time-dependent rules are tested with a frozen clock.

Coverage targets: at least 70% line and 60% branch coverage of backend code, enforced in CI (NFR-MNT-01); every requirement covered by at least one test case (Section 13).

Defect management: defects are GitHub Issues labelled bug and severity (S1 blocker, S2 major, S3 minor, S4 cosmetic), linked to the failing test case and placed on the GitHub Projects board. S1 and S2 defects must be fixed before a deliverable is submitted.

Entry criteria: the build passes CI; test data is loaded; the features under test are merged.

Exit criteria: all High priority test cases pass; no open S1 or S2 defects; coverage targets met; the traceability matrix shows every requirement covered.

5.1 Security Validation

Each security requirement from SRS Section 6.3 is validated by the method below. Automated checks run in CI; scans run before each deliverable.

Requirement	Validation method	Tool	Test cases
SEC-01 password hashing and policy	Inspect stored hash format; try weak passwords	pytest	TC-39
SEC-02 lockout and rate limit	Scripted failed sign-ins; burst of requests from one IP	pytest, Locust	TC-03
SEC-03 server-side role checks	Call every endpoint as every role and compare with the role matrix (SRS B.3)	pytest (parametrised)	TC-15, TC-21
SEC-04 HTTPS and cookies	Check redirect, TLS version and cookie flags on staging	testssl.sh, browser dev tools	TC-38
SEC-05 CSRF	Send state-changing requests without a token	pytest	TC-16
SEC-06 input validation, injection	Malicious input in every form; active scan	pytest, OWASP ZAP, bandit (static analysis)	TC-17
SEC-07 audit log	Perform each privileged action; check entries; try to modify them	pytest, psql	TC-19, TC-31
SEC-08 idle timeout, session rotation	Compare session ids before and after sign-in; wait out the timeout	Playwright	TC-38
SEC-09 masking, no secrets in logs	Inspect attendant screen; grep logs after a test run	Playwright, grep	TC-22, TC-39
SEC-10 secrets outside the repository	Commit a fake secret on a test branch	gitleaks in CI	TC-40

In addition, pip-audit checks dependencies for known vulnerabilities on every pull request, and an OWASP ZAP baseline scan of the staging site is run before each deliverable. Findings rated Medium or higher are logged as S2 defects.

6. Test Environment

- **Developer machines:** Linux, macOS or Windows with Python 3.12 and PostgreSQL 16 (local or Docker).
- **CI:** GitHub Actions, ubuntu-latest runner with a PostgreSQL 16 service container.
- **Staging:** one Linux server (2 vCPU, 4 GB RAM) running nginx, Gunicorn, PostgreSQL 16 and the forecast service, used for system, performance and security tests.

- **Browsers:** latest two versions of Chrome, Firefox, Edge and Safari; phone viewport 360 pixels wide.
- **Tools:** pytest, pytest-django, pytest-cov, freezegun, Playwright, Locust, OWASP ZAP, bandit, pip-audit, gitleaks, axe-core, testssl.sh.

7. Test Schedule

Milestone	Date	Notes
Test plan and test case design (this document)	07-Oct-2026	Part-1 deliverable
CI with unit and integration suites running	With the first implementation sprint	Suites grow with each feature
Validation specification update after design	Per course calendar	Date to be confirmed when announced
System test execution and security scans	Before the system validation deliverable	Date to be confirmed
Acceptance sessions and final regression	Before the final demo	Date to be confirmed

Only the first date is fixed; the others follow the course deliverable calendar and will be filled in when it is published.

8. Test Deliverables

- Test plan (this document) and its updates.
- Test cases (Section 16) and the automated test code in the tests/ folder of the repository.
- Test data factories and the synthetic data generator.
- CI run logs and coverage reports (line and branch) for every pull request.
- Defect reports as GitHub Issues.
- Security scan reports (ZAP, bandit, pip-audit, gitleaks).
- System validation report and test summary for the final deliverable.

9. Roles and Responsibilities

Role	Name	Responsibility
Test lead, CI and security scans	Vivian Sobers E (Integration and DevOps)	Owens this plan, the CI pipeline, performance tests, ZAP and secret scanning; integration of the forecast service
Data and integration tests	D Harikrishnan (Database)	Constraint, concurrency and migration tests; seeded data sets; report figures
UI and system tests	Dhruv Purushotham Raju (Frontend)	Playwright flows, cross-browser, accessibility and usability sessions
Unit tests and defect fixing	Gitanjali A (Backend)	Unit tests for services, API tests, fixing backend defects

Every developer writes unit tests for their own code; the leads above own the shared suites.

10. Risks and Mitigation

Risk	Mitigation
Concurrency bugs that appear only under load	Repeat TC-07 20 times per run; run it in CI against real PostgreSQL, not SQLite
Time-dependent tests that fail at random	Freeze the clock in every test that involves expiry, windows or billing
No staging server early in the semester	Run system tests against a local Docker setup until staging exists
Team members new to testing tools	Shared examples for each tool in the tests/ folder; pair on the first tests
Late features leave little time for testing	Tests are written in the same pull request as the feature; CI blocks merges without them

11. Assumptions & Dependencies

- The SRS requirements are stable for this release; changes go through a pull request that updates the SRS, this plan and the traceability matrix together.
- GitHub Actions minutes are sufficient for the suites (the repository is public).
- A test SMTP inbox (for example a local mail catcher) is available for e-mail tests.
- Five volunteers are available for the usability sessions (TC-41).

12. Suspension & Resumption Criteria

Suspend testing if: CI is broken on main for more than 4 hours; the staging environment is unavailable for more than 4 hours; or an S1 defect blocks more than 30% of the test cases.

Resume testing when: the blocking defect is fixed and verified; the environment is restored; and a smoke test (TC-01, TC-06, TC-13) passes.

13. Test Case Management & Traceability

Test cases are identified TC-01 to TC-43 and specified in Section 16. Each automated test function names its test case in its docstring, and each test case lists the SRS requirements it verifies. The traceability matrix below is the reverse view: for every requirement, the test cases that verify it. The same mapping appears in Appendix C of the SRS.

Coverage: 71 of 72 requirements have at least one test case. The remaining one, NFR-AVL-01, is verified by uptime monitoring (Section 14).

Requirement	Description	Test cases
FR-AUTH-01	Driver self-registration	TC-01, TC-02
FR-AUTH-02	Sign in and sign out	TC-01, TC-03
FR-AUTH-03	Three roles; staff created by admin	TC-21

Requirement	Description	Test cases
FR-AUTH-04	Password reset link	TC-22
FR-AUTH-05	Deactivate staff accounts	TC-21
FR-AUTH-06	Accessible-parking permit flag	TC-23
FR-VEH-01	Up to 5 vehicles per Driver	TC-24
FR-VEH-02	Registration number format	TC-04
FR-VEH-03	Unique registration number	TC-04
FR-VEH-04	Protect vehicles with bookings	TC-24
FR-LOT-01	Manage lots	TC-25
FR-LOT-02	Manage slots	TC-25
FR-LOT-03	Block slot for maintenance	TC-05
FR-LOT-04	Move bookings off blocked slot	TC-26
FR-SRCH-01	Availability search	TC-05
FR-SRCH-02	Search window validation	TC-27
FR-SRCH-03	Live occupancy view	TC-28
FR-RES-01	Create reservation	TC-06
FR-RES-02	No double booking	TC-07
FR-RES-03	Reservation limit and unpaid charges	TC-08
FR-RES-04	Cancellation and late fee	TC-09
FR-RES-05	Expiry and no-show fee	TC-10
FR-RES-06	Booking history	TC-29
FR-RES-07	Booking e-mails	TC-10, TC-26
FR-ALLOC-01	Slot eligibility rules	TC-06, TC-23
FR-ALLOC-02	Deterministic slot choice	TC-06
FR-ALLOC-03	Retry on conflict	TC-07
FR-GATE-01	Check-in window	TC-11
FR-GATE-02	Early arrival refused	TC-11
FR-GATE-03	Walk-in registration	TC-12
FR-GATE-04	Check-out and receipt	TC-13
FR-GATE-05	Record payment	TC-30
FR-GATE-06	Manual time correction	TC-31
FR-GATE-07	Reallocate occupied slot at check-in	TC-32
FR-BILL-01	Tariff configuration	TC-33
FR-BILL-02	Parking charge calculation	TC-13
FR-BILL-03	Overstay charge	TC-14
FR-BILL-04	Receipt contents	TC-13
FR-BILL-05	Unpaid charges shown	TC-30
FR-RPT-01	Live occupancy dashboard	TC-28
FR-RPT-02	Occupancy report	TC-34
FR-RPT-03	Revenue report	TC-34
FR-RPT-04	Audit log search	TC-19
FR-FCST-01	Availability forecast	TC-35
FR-FCST-02	Weekly retraining and baseline	TC-35

Requirement	Description	Test cases
FR-FCST-03	Forecast fallback	TC-20
NFR-PERF-01	Search response time	TC-18
NFR-PERF-02	Booking and gate response time	TC-36
NFR-PERF-03	200 concurrent sessions	TC-18
NFR-PERF-04	Page load time and size	TC-37
NFR-PERF-05	Forecast response time	TC-36
SAF-01	Blocked slots never allocated	TC-05
SAF-02	Accessible slots protected	TC-23
SAF-03	Manual fallback at the gate	TC-31
SEC-01	Password hashing and policy	TC-39
SEC-02	Lockout and rate limit	TC-03
SEC-03	Server-side role checks	TC-15
SEC-04	HTTPS and secure cookies	TC-38
SEC-05	CSRF protection	TC-16
SEC-06	Input validation, parameterised queries	TC-17
SEC-07	Audit log	TC-19, TC-31
SEC-08	Idle timeout, session rotation	TC-38
SEC-09	Masking; no secrets on screen or in logs	TC-22, TC-39
SEC-10	Secrets outside the repository	TC-40
NFR-REL-01	Exactly one winner for last slot	TC-07
NFR-AVL-01	99% availability in operating hours	Uptime monitoring (Section 14)
NFR-USE-01	First booking under 3 minutes	TC-41
NFR-USE-02	Small screens and contrast	TC-42
NFR-MNT-01	Test coverage thresholds	TC-43
NFR-MNT-02	Lint and reviewed merges	TC-43
NFR-PRT-01	Server and browser support	TC-42
NFR-TST-01	Rules testable without web layer	TC-43

14. Test Metrics & Reporting

- Test cases executed, passed, failed and blocked, by level and priority.
- Requirement coverage: requirements with at least one passing test case.
- Code coverage: line and branch, per module, from CI.
- Defects by severity, open and closed, and defect age.
- Performance: 95th percentile response time per endpoint against NFR-PERF targets.
- Availability: monthly uptime between 06:00 and 23:00 IST from the uptime monitor (NFR-AVL-01).

Reports: a CI summary on every pull request; a short status note in each sprint review on the GitHub Projects board; and a final test summary report with the system validation deliverable.

15. Approvals

Role	Name	Signature / Date
Test lead	Vivian Sobers E	
Database lead	D Harikrishnan	
Frontend lead	Dhruv Purushotham Raju	
Backend lead	Gitanjali A	
Course instructor		

16. Test Cases

43 test cases cover functional requirements (27), security (9) and non-functional requirements (7). None has been executed yet; the Status field will be filled in when they are run.

ID	Title	Requirements	Type	Level	Priority
TC-01	Register a Driver with valid details	FR-AUTH-01, FR-AUTH-02	Functional	System	High
TC-02	Reject duplicate e-mail at registration	FR-AUTH-01	Functional	Integration	High
TC-03	Lock account after 5 failed sign-ins	SEC-02, FR-AUTH-02	Security	Integration	High
TC-04	Validate vehicle registration number format	FR-VEH-02, FR-VEH-03	Functional	Unit	Medium
TC-05	Search excludes blocked and occupied slots	FR-SRCH-01, FR-LOT-03, SAF-01	Functional	Integration	High
TC-06	Allocate the preferred, nearest slot	FR-ALLOC-01, FR-ALLOC-02, FR-RES-01	Functional	Unit	High
TC-07	Exactly one booking wins the last slot	FR-RES-02, FR-ALLOC-03, NFR-REL-01	Non-functional (reliability)	Integration	High
TC-08	Enforce reservation limit and unpaid charges	FR-RES-03	Functional	Integration	Medium
TC-09	Late cancellation records a fee	FR-RES-04	Functional	Unit	Medium
TC-10	Unclaimed reservation expires	FR-RES-05, FR-RES-07	Functional	Integration	High
TC-11	Check-in window is enforced	FR-GATE-01, FR-GATE-02	Functional	System	High
TC-12	Register a walk-in vehicle	FR-GATE-03	Functional	System	Medium
TC-13	Compute parking charge with rounding and minimum	FR-BILL-02, FR-GATE-04, FR-BILL-04	Functional	Unit	High
TC-14	Bill overstay at 1.5 times the rate	FR-BILL-03	Functional	Unit	High
TC-15	Attendant cannot change a tariff	SEC-03, BR-01	Security	Integration	High
TC-16	State change without CSRF token is rejected	SEC-05	Security	Integration	High
TC-17	SQL injection input is handled safely	SEC-06	Security	System	High
TC-18	Search meets the response-time target	NFR-PERF-01, NFR-PERF-03	Non-functional (performance)	System	Medium
TC-19	Privileged actions are audited	SEC-07, FR-RPT-04	Security	Integration	Medium
TC-20	Search works when the forecast service is down	FR-FCST-03	Functional	Integration	Low
TC-21	Only administrators create staff; deactivation blocks sign-in	FR-AUTH-03, FR-AUTH-05	Functional	Integration	High
TC-22	Password reset link is single-use and expires	FR-AUTH-04, SEC-09	Security	Integration	Medium
TC-23	Accessible slots only for permit holders	FR-AUTH-06, FR-ALLOC-01, SAF-02	Functional	Unit	High
TC-24	Vehicle limit and protected removal	FR-VEH-01, FR-VEH-04	Functional	Unit	Medium
TC-25	Create lots and slots with unique codes	FR-LOT-01, FR-LOT-02	Functional	Integration	Medium
TC-26	Blocking a slot moves affected reservations	FR-LOT-04, FR-RES-07	Functional	Integration	High

ID	Title	Requirements	Type	Level	Priority
TC-27	Search window validation	FR-SRCH-02	Functional	Unit	Medium
TC-28	Live occupancy view refreshes	FR-SRCH-03, FR-RPT-01	Functional	System	Medium
TC-29	Booking history is paged and ordered	FR-RES-06	Functional	Integration	Low
TC-30	Record a payment and clear unpaid charges	FR-GATE-05, FR-BILL-05, BR-04	Functional	Integration	High
TC-31	Manual time correction is limited and audited	FR-GATE-06, SAF-03, SEC-07	Functional	Integration	Medium
TC-32	Reallocate when the reserved slot is still occupied	FR-GATE-07	Functional	Integration	Medium
TC-33	Tariff in force at entry is used	FR-BILL-01, BR-08	Functional	Unit	High
TC-34	Occupancy and revenue reports export to CSV	FR-RPT-02, FR-RPT-03	Functional	Integration	Low
TC-35	Forecast bands and baseline label	FR-FCST-01, FR-FCST-02	Functional	Unit	Low
TC-36	Booking, gate and forecast response times	NFR-PERF-02, NFR-PERF-05	Non-functional (performance)	System	Medium
TC-37	Page weight and load time	NFR-PERF-04	Non-functional (performance)	System	Low
TC-38	TLS, cookie flags and idle timeout	SEC-04, SEC-08	Security	System	High
TC-39	Password storage and data masking	SEC-01, SEC-09	Security	Integration	High
TC-40	No secrets in the repository	SEC-10	Security	System	Medium
TC-41	First-time booking within 3 minutes	NFR-USE-01	Non-functional (usability)	Acceptance	Medium
TC-42	Small screens and colour contrast	NFR-USE-02, NFR-PRT-01	Non-functional (usability)	System	Low
TC-43	Coverage and lint gates in CI	NFR-MNT-01, NFR-MNT-02, NFR-TST-01	Non-functional (maintainability)	System	Medium

TC-01: Register a Driver with valid details

Field	Detail
Requirements	FR-AUTH-01, FR-AUTH-02
Type / level / priority	Functional / System / High
Preconditions	No account exists for asha@example.com.
Steps	1. Open the registration page. 2. Enter name Asha Rao, e-mail asha@example.com, mobile 9876543210, password Park2026. 3. Submit.
Expected result	Account created with role Driver; user is signed in and sees the Driver home page.
Status	Not yet executed

TC-02: Reject duplicate e-mail at registration

Field	Detail
Requirements	FR-AUTH-01
Type / level / priority	Functional / Integration / High
Preconditions	An account exists for asha@example.com.
Steps	1. Register again with e-mail ASHA@example.com and otherwise valid data.
Expected result	Registration refused with "An account with this e-mail already exists"; no new account created.
Status	Not yet executed

TC-03: Lock account after 5 failed sign-ins

Field	Detail
Requirements	SEC-02, FR-AUTH-02
Type / level / priority	Security / Integration / High
Preconditions	Active Driver account asha@example.com.
Steps	1. Submit 5 sign-ins with a wrong password within 2 minutes. 2. Submit a sixth sign-in with the correct password.
Expected result	Attempts 1 to 5 return "Invalid e-mail or password"; attempt 6 returns HTTP 423 with the unlock time; audit log has 5 auth.failed entries.
Status	Not yet executed

TC-04: Validate vehicle registration number format

Field	Detail
Requirements	FR-VEH-02, FR-VEH-03
Type / level / priority	Functional / Unit / Medium
Preconditions	Signed in as a Driver with no vehicles.
Steps	1. Add vehicle "ka 01 ab 1234" (Car). 2. Add vehicle "22BH1234AA" (Car). 3. Add vehicle "KA01AB12" (Car). 4. Add "KA01AB1234" again.
Expected result	First stored as KA01AB1234; second accepted; third rejected as invalid format; fourth rejected as duplicate.
Status	Not yet executed

TC-05: Search excludes blocked and occupied slots

Field	Detail
Requirements	FR-SRCH-01, FR-LOT-03, SAF-01
Type / level / priority	Functional / Integration / High
Preconditions	Lot A has 4 Standard slots: S1 free, S2 blocked for maintenance, S3 reserved 10:00 to 12:00, S4 free.
Steps	1. Search Lot A for a Car from 11:00 to 13:00.
Expected result	Result shows 2 eligible free slots (S1, S4).
Status	Not yet executed

TC-06: Allocate the preferred, nearest slot

Field	Detail
Requirements	FR-ALLOC-01, FR-ALLOC-02, FR-RES-01
Type / level / priority	Functional / Unit / High
Preconditions	Lot A free slots for 10:00 to 12:00: EV1 (rank 5), S2 (rank 1), S1 (rank 1). Two Drivers, each with an Electric car.
Steps	1. Driver 1 reserves 10:00 to 12:00. 2. Driver 2 reserves 10:00 to 12:00.
Expected result	Driver 1 gets EV1 (EV charging preferred over the nearer Standard slots). Driver 2 gets S1 (no EV slot left; lowest rank, then lowest code).
Status	Not yet executed

TC-07: Exactly one booking wins the last slot

Field	Detail
Requirements	FR-RES-02, FR-ALLOC-03, NFR-REL-01
Type / level / priority	Non-functional (reliability) / Integration / High
Preconditions	Lot B has exactly 1 eligible free slot for 10:00 to 11:00; 50 Driver accounts with Cars.
Steps	1. Send 50 reservation requests for that window at the same time (thread pool against the API). 2. Repeat the run 20 times.
Expected result	Every run creates exactly 1 reservation; 49 requests receive "No slot available for this window"; the database never holds overlapping active reservations for the slot.
Status	Not yet executed

TC-08: Enforce reservation limit and unpaid charges

Field	Detail
Requirements	FR-RES-03
Type / level / priority	Functional / Integration / Medium
Preconditions	Driver has 2 Reserved reservations.
Steps	1. Try a third reservation. 2. Cancel one, record an unpaid no-show charge for the Driver, try again.
Expected result	First attempt refused with the reservation-limit reason; second refused with the unpaid-charge reason.
Status	Not yet executed

TC-09: Late cancellation records a fee

Field	Detail
Requirements	FR-RES-04
Type / level / priority	Functional / Unit / Medium
Preconditions	Reservations R1 starting in 90 minutes and R2 starting in 30 minutes.
Steps	1. Cancel R1. 2. Cancel R2.
Expected result	R1 Cancelled with no charge; R2 Cancelled with a Rs 30.00 late-cancellation charge.
Status	Not yet executed

TC-10: Unclaimed reservation expires

Field	Detail
Requirements	FR-RES-05, FR-RES-07
Type / level / priority	Functional / Integration / High
Preconditions	Reservation starting at 10:00, not checked in; clock controlled in the test.
Steps	1. Advance the clock to 10:14 and run the expiry job. 2. Advance to 10:16 and run it again.
Expected result	Still Reserved at 10:14; Expired at 10:16 with slot freed, Rs 50.00 no-show charge and an expiry e-mail queued.
Status	Not yet executed

TC-11: Check-in window is enforced

Field	Detail
Requirements	FR-GATE-01, FR-GATE-02
Type / level / priority	Functional / System / High
Preconditions	Reservation starting at 10:00.
Steps	1. Attendant checks in at 09:40. 2. Attendant checks in at 09:50.
Expected result	09:40 refused with "Check-in opens at 09:45"; 09:50 accepted, state Active, entry time 09:50.
Status	Not yet executed

TC-12: Register a walk-in vehicle

Field	Detail
Requirements	FR-GATE-03
Type / level / priority	Functional / System / Medium
Preconditions	Lot A has one Standard slot free until closing time and another reserved from 2 hours from now; signed in as Gate Attendant.
Steps	1. Register walk-in KA05MN4321 as Car.
Expected result	Active walk-in reservation created on the slot that is free until closing, with a booking reference; the slot shows as occupied.
Status	Not yet executed

TC-13: Compute parking charge with rounding and minimum

Field	Detail
Requirements	FR-BILL-02, FR-GATE-04, FR-BILL-04
Type / level / priority	Functional / Unit / High
Preconditions	Tariff Rs 40 per hour for Standard slots; reservations ending well after exit.
Steps	1. Check out stays of 20 min, 61 min and 2 h 10 min.
Expected result	Charges Rs 40.00 (1 h minimum), Rs 60.00 (1.5 h), Rs 100.00 (2.5 h); receipts list every field in FR-BILL-04.
Status	Not yet executed

TC-14: Bill overstay at 1.5 times the rate

Field	Detail
Requirements	FR-BILL-03
Type / level / priority	Functional / Unit / High
Preconditions	Tariff Rs 40 per hour; reservation 10:00 to 12:00; entry 10:00.
Steps	1. Check out at 12:08. 2. For a second identical reservation, check out at 12:40.
Expected result	12:08: Rs 80.00, no overstay (within grace). 12:40: Rs 80.00 parking plus Rs 30.00 overstay (30 min at Rs 60 per hour).
Status	Not yet executed

TC-15: Attendant cannot change a tariff

Field	Detail
Requirements	SEC-03, BR-01
Type / level / priority	Security / Integration / High
Preconditions	Signed in as Gate Attendant.
Steps	1. Send POST /api/v1/tariffs directly with a valid CSRF token and a new rate.
Expected result	HTTP 403; tariff unchanged; the denied attempt is logged.
Status	Not yet executed

TC-16: State change without CSRF token is rejected

Field	Detail
Requirements	SEC-05
Type / level / priority	Security / Integration / High
Preconditions	Signed in as Driver with a Reserved reservation.
Steps	1. Send POST /api/v1/reservations/{ref}/cancel without a CSRF token.
Expected result	HTTP 403; reservation still Reserved.
Status	Not yet executed

TC-17: SQL injection input is handled safely

Field	Detail
Requirements	SEC-06
Type / level / priority	Security / System / High
Preconditions	Signed in as Driver.
Steps	1. Search with lot parameter "1 OR 1=1" and add a vehicle named "x'); DROP TABLE vehicle;--". 2. Run an OWASP ZAP active scan on the search and vehicle endpoints.
Expected result	Search returns HTTP 400 for the invalid lot; the nickname is stored as plain text; no SQL error appears; ZAP reports no SQL injection findings.
Status	Not yet executed

TC-18: Search meets the response-time target

Field	Detail
Requirements	NFR-PERF-01, NFR-PERF-03
Type / level / priority	Non-functional (performance) / System / Medium
Preconditions	Staging database with one lot of 500 slots and 30 days of synthetic reservations.
Steps	1. Run a Locust test with 50 users searching continuously for 10 minutes.
Expected result	95th percentile response time at or below 1 second; error rate 0%.
Status	Not yet executed

TC-19: Privileged actions are audited

Field	Detail
Requirements	SEC-07, FR-RPT-04
Type / level / priority	Security / Integration / Medium
Preconditions	Signed in as Lot Administrator.
Steps	1. Change the Standard tariff from Rs 40 to Rs 50. 2. Search the audit log for action tariff.updated. 3. Try to delete the entry through the application and the API.
Expected result	Entry shows actor, old value 40.00, new value 50.00 and timestamp; delete attempts fail with HTTP 403 or 405.
Status	Not yet executed

TC-20: Search works when the forecast service is down

Field	Detail
Requirements	FR-FCST-03
Type / level / priority	Functional / Integration / Low
Preconditions	Forecast service stopped.
Steps	1. Run an availability search with a future arrival time.
Expected result	Search result appears within 3 seconds; forecast panel shows "Forecast not available".
Status	Not yet executed

TC-21: Only administrators create staff; deactivation blocks sign-in

Field	Detail
Requirements	FR-AUTH-03, FR-AUTH-05
Type / level / priority	Functional / Integration / High
Preconditions	Lot Administrator and Gate Attendant accounts exist; the attendant is signed in on another browser.
Steps	1. As the attendant, try to create a Gate Attendant account. 2. As the administrator, create one, then deactivate the signed-in attendant. 3. Wait 1 minute and reload the attendant's page; try to sign in again.
Expected result	Attendant's attempt gets HTTP 403. Administrator's succeeds. The attendant's session has ended and sign-in is refused.
Status	Not yet executed

TC-22: Password reset link is single-use and expires

Field	Detail
Requirements	FR-AUTH-04, SEC-09
Type / level / priority	Security / Integration / Medium
Preconditions	Driver account; mail sent to a test inbox.
Steps	1. Request a reset and use the link to set a new password. 2. Open the same link again. 3. Request another reset and open the link 31 minutes later. 4. Search application logs for the token.
Expected result	First use succeeds; second use and the expired link are refused; the token never appears in logs.
Status	Not yet executed

TC-23: Accessible slots only for permit holders

Field	Detail
Requirements	FR-AUTH-06, FR-ALLOC-01, SAF-02
Type / level / priority	Functional / Unit / High
Preconditions	Lot C has only Accessible slots free. Driver P holds a permit; Driver Q does not.
Steps	1. Driver Q searches and reserves for a Car. 2. Driver P does the same.
Expected result	Driver Q sees 0 eligible slots and cannot reserve; Driver P is allocated an Accessible slot.
Status	Not yet executed

TC-24: Vehicle limit and protected removal

Field	Detail
Requirements	FR-VEH-01, FR-VEH-04
Type / level / priority	Functional / Unit / Medium
Preconditions	Driver has 5 vehicles; one has a Reserved reservation.
Steps	1. Add a sixth vehicle. 2. Remove the vehicle with the reservation. 3. Remove a vehicle with no reservation.
Expected result	Sixth vehicle refused; removal of the reserved vehicle refused with the reason; the other removal succeeds.
Status	Not yet executed

TC-25: Create lots and slots with unique codes

Field	Detail
Requirements	FR-LOT-01, FR-LOT-02
Type / level / priority	Functional / Integration / Medium
Preconditions	Signed in as Lot Administrator.
Steps	1. Create Lot D, open 06:00 to 23:00. 2. Add slot D-01 (Standard, rank 1). 3. Add D-01 again. 4. Add slot with rank 0.
Expected result	Lot and first slot created; duplicate code and rank 0 are rejected with field messages.
Status	Not yet executed

TC-26: Blocking a slot moves affected reservations

Field	Detail
Requirements	FR-LOT-04, FR-RES-07
Type / level / priority	Functional / Integration / High
Preconditions	Slot S1 has a Reserved booking 14:00 to 16:00; S4 is free for that window; S5 is not.
Steps	1. Block S1 from 13:00 to 17:00. 2. Then block S4 for the same period.
Expected result	The booking moves to S4 and an e-mail with the new slot is queued; after S4 is blocked the booking is cancelled with no fee and the Driver is e-mailed.
Status	Not yet executed

TC-27: Search window validation

Field	Detail
Requirements	FR-SRCH-02
Type / level / priority	Functional / Unit / Medium
Preconditions	Lot A open 06:00 to 23:00.
Steps	1. Search windows: starting 10 minutes ago; starting 8 days ahead; 20 minutes long; 25 hours long; 22:00 to 23:30.
Expected result	Each is refused with a message naming the broken rule.
Status	Not yet executed

TC-28: Live occupancy view refreshes

Field	Detail
Requirements	FR-SRCH-03, FR-RPT-01
Type / level / priority	Functional / System / Medium
Preconditions	Occupancy page open in a browser; attendant on a second browser.
Steps	1. Attendant checks a vehicle in. 2. Watch the occupancy page for 30 seconds.
Expected result	Counts change within 30 seconds without a manual reload; the administrator dashboard shows the same counts.
Status	Not yet executed

TC-29: Booking history is paged and ordered

Field	Detail
Requirements	FR-RES-06
Type / level / priority	Functional / Integration / Low
Preconditions	Driver with 45 past and current reservations.
Steps	1. Open booking history and go through all pages.
Expected result	3 pages of 20, 20 and 5 entries, newest first, each showing the fields in FR-RES-06.
Status	Not yet executed

TC-30: Record a payment and clear unpaid charges

Field	Detail
Requirements	FR-GATE-05, FR-BILL-05, BR-04
Type / level / priority	Functional / Integration / High
Preconditions	Driver has an unpaid Rs 50.00 no-show charge.
Steps	1. Check the Driver's home page. 2. Attendant records a UPI payment for the charge. 3. Driver makes a new reservation.
Expected result	Home page shows Rs 50.00 unpaid; after payment it shows Rs 0.00 and the payment appears in the audit log; the new reservation succeeds.
Status	Not yet executed

TC-31: Manual time correction is limited and audited

Field	Detail
Requirements	FR-GATE-06, SAF-03, SEC-07
Type / level / priority	Functional / Integration / Medium
Preconditions	Active reservation; signed in as Gate Attendant.
Steps	1. Enter an exit time 2 hours ago with reason "system outage". 2. Enter an exit time 25 hours ago. 3. Enter a correction with no reason.
Expected result	First accepted and audited with the reason; second and third refused.
Status	Not yet executed

TC-32: Reallocate when the reserved slot is still occupied

Field	Detail
Requirements	FR-GATE-07
Type / level / priority	Functional / Integration / Medium
Preconditions	Reservation R for slot S1 at 12:00; the previous vehicle in S1 has not checked out; S2 free.
Steps	1. Attendant checks in R at 12:00.
Expected result	R becomes Active on S2 and the new slot code is shown.
Status	Not yet executed

TC-33: Tariff in force at entry is used

Field	Detail
Requirements	FR-BILL-01, BR-08
Type / level / priority	Functional / Unit / High
Preconditions	Standard tariff Rs 40 per hour; a vehicle entered yesterday and is still parked.
Steps	1. Set a new rate of Rs 50 effective today. 2. Check out the parked vehicle after 2 hours of billable time.
Expected result	Charge uses Rs 40 per hour (Rs 80.00); a vehicle entering today is billed at Rs 50.
Status	Not yet executed

TC-34: Occupancy and revenue reports export to CSV

Field	Detail
Requirements	FR-RPT-02, FR-RPT-03
Type / level / priority	Functional / Integration / Low
Preconditions	Seeded data for 7 days with known totals.
Steps	1. Run both reports for the 7 days and download the CSV files. 2. Ask for a 32-day range.
Expected result	Figures match the seeded totals and the CSV columns match Appendix B of the SRS; the 32-day range is refused.
Status	Not yet executed

TC-35: Forecast bands and baseline label

Field	Detail
Requirements	FR-FCST-01, FR-FCST-02
Type / level / priority	Functional / Unit / Low
Preconditions	Forecast service with a stub model returning 0.35, 0.60 and 0.90; less than 4 weeks of history.
Steps	1. Request forecasts for three arrival times.
Expected result	Shown as 35% Low, 60% Medium, 90% High, each labelled "Estimate".
Status	Not yet executed

TC-36: Booking, gate and forecast response times

Field	Detail
Requirements	NFR-PERF-02, NFR-PERF-05
Type / level / priority	Non-functional (performance) / System / Medium
Preconditions	Staging data as in TC-18.
Steps	1. Run a Locust test with 50 users mixing reservation, check-in and check-out for 10 minutes. 2. Send 1,000 forecast requests.
Expected result	95th percentile at or below 2 seconds for reservation and gate actions and at or below 500 ms for forecasts.
Status	Not yet executed

TC-37: Page weight and load time

Field	Detail
Requirements	NFR-PERF-04
Type / level / priority	Non-functional (performance) / System / Low
Preconditions	Chrome with network throttled to 10 Mbps.
Steps	1. Load the search, booking and receipt pages with an empty cache.
Expected result	Each page loads within 3 seconds and transfers 500 KB or less.
Status	Not yet executed

TC-38: TLS, cookie flags and idle timeout

Field	Detail
Requirements	SEC-04, SEC-08
Type / level / priority	Security / System / High
Preconditions	Staging deployment behind nginx.
Steps	1. Request the site over plain HTTP. 2. Inspect the TLS version and the session cookie after sign-in. 3. Stay idle for 31 minutes, then reload.
Expected result	HTTP redirects to HTTPS; TLS 1.2 or later; cookie is Secure, HttpOnly, SameSite=Lax; session id changes at sign-in; after 31 minutes the user must sign in again.
Status	Not yet executed

TC-39: Password storage and data masking

Field	Detail
Requirements	SEC-01, SEC-09
Type / level / priority	Security / Integration / High
Preconditions	Driver registered with password Park2026 and mobile 9876543210.
Steps	1. Read the user row in the database. 2. Register with password "password". 3. View the Driver on the attendant screen.
Expected result	Stored value starts with pbkdf2_sha256\$; the weak password is refused; the attendant sees 3210 .
Status	Not yet executed

TC-40: No secrets in the repository

Field	Detail
Requirements	SEC-10
Type / level / priority	Security / System / Medium
Preconditions	CI pipeline with a secret scanner (gitLeaks).
Steps	1. Push a branch that adds a fake SECRET_KEY value to settings.py.
Expected result	CI fails at the secret scan and the pull request cannot be merged.
Status	Not yet executed

TC-41: First-time booking within 3 minutes

Field	Detail
Requirements	NFR-USE-01
Type / level / priority	Non-functional (usability) / Acceptance / Medium
Preconditions	5 classmates who have not used the system.
Steps	1. Ask each to add a vehicle and book a slot without help; time them.
Expected result	All 5 finish in under 3 minutes using no more than 4 booking screens.
Status	Not yet executed

TC-42: Small screens and colour contrast

Field	Detail
Requirements	NFR-USE-02, NFR-PRT-01
Type / level / priority	Non-functional (usability) / System / Low
Preconditions	Playwright with Chrome, Firefox, Edge and WebKit.
Steps	1. Run the booking flow at 360 px width in each browser. 2. Run axe-core contrast checks on every page.
Expected result	Flow completes in all browsers with no horizontal scrolling; no contrast violations at level AA.
Status	Not yet executed

TC-43: Coverage and lint gates in CI

Field	Detail
Requirements	NFR-MNT-01, NFR-MNT-02, NFR-TST-01
Type / level / priority	Non-functional (maintainability) / System / Medium
Preconditions	CI configured with coverage thresholds; branch protection on main.
Steps	1. Open a pull request that lowers coverage below 70% line or 60% branch. 2. Open one with a ruff error. 3. Try to push directly to main.
Expected result	Both pull requests fail CI and cannot merge; the direct push is rejected; allocation and billing unit tests run without the web layer.
Status	Not yet executed